COSC 3337 – **Music Data Project**

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1. Data Collection & Preprocessing

1.1. Data Collection

For this project, we collected data from Spotify, which provides a comprehensive record of users' streaming history, from the first time they listened to music on the platform. The data can be downloaded from Spotify's privacy settings page, where users can request a copy of their data. This data comes in the form of a zip file containing multiple JSON files for each year of listening history. Each JSON file contains an array of listening events, with attributes such as timestamp, track name, artist name, album name, platform, country, and more.

For example, a single listening event in the JSON file looks like this:

```json
{
  "ts": "2026-02-02T01:15:20Z",
  "platform": "windows",
  "ms_played": 296413,
  "conn_country": "US",
  "ip_addr": "129.7.0.128",
  "master_metadata_track_name": "ラストナイト -album version-",
  "master_metadata_album_artist_name": "toe",
  "master_metadata_album_album_name": "For Long Tomorrow",
  "spotify_track_uri": "spotify:track:0Hi2e0tK8xefutVU3oE9JG",
  "episode_name": null,
  "episode_show_name": null,
  "spotify_episode_uri": null,
  "audiobook_title": null,
  "audiobook_uri": null,
  "audiobook_chapter_uri": null,
  "audiobook_chapter_title": null,
  "reason_start": "trackdone",
  "reason_end": "trackdone",
  "shuffle": true,
  "skipped": false,
  "offline": false,
  "offline_timestamp": 1769994623,
  "incognito_mode": false
},
```

where each attribute provides specific information about the listening event, such as the timestamp of when the track was played, the platform used, how long the track was played for in milliseconds, the country where it was played, and details about the track itself (name, artist, album). Additionally, there are attributes that indicate whether the user was shuffling their music, whether they skipped the track, and whether they were listening offline.

We collected data from four users over a span of 11 years (2015-2026), in the form of 58 total JSON files. Some users have more extensive listening histories than others, with varying numbers of unique tracks, artists, albums, platforms, and countries represented in their data. However, there is a full overlap in the time span of their listening history from 2020 to 2026.

1.2. Data Preprocessing

To prepare the data for analysis, we performed several preprocessing steps. First, we extracted the relevant attributes from the JSON files and gave them more descriptive names. The data from Spotify contained some attributes that were not relevant to our analysis, such as IP address and audiobook-related fields, which we discarded. We focused on attributes that were relevant to understanding listening habits, such as track name, artist name, album name, platform, country, timestamp, ms played, reason for starting and ending the track, shuffle status, skip status, and offline status. We then imported the data into a central data store, using Supabase, which is a PostgreSQL database. This allowed us to easily query and manipulate the data for our analysis.

The schema we used for the database is as follows:

```sql
create table public.listening_history (
  track_id text not null,
  timestamp timestamp without time zone not null,
  username text not null,
  track_name text not null,
  artist_name text not null,
  album_name text null,
  platform text null,
  ms_played bigint null,
  conn_country text null,
  reason_start text null,
  reason_end text null,
  shuffle boolean null,
  skipped boolean null,
  offline boolean null,
  offline_timestamp text null,
  constraint spotify_data_pkey primary key (track_id, "timestamp", username)
);
```

The primary key is a combination of `track_id`, `timestamp`, and `username`, which ensures that each listening event is uniquely identified. This also allows us to easily compare data from multiple users while maintaining the integrity of the data.

We also pruned any null entries and duplicate songs (based on the primary key of username and timestamp) to ensure data quality. To make sure we had a clean dataset, we removed any songs that had both 1 play and less than 30 seconds played, as these were likely accidental plays or songs that were not fully listened to. After pruning, we were left with approximately 845,000 listening events. Additionally, we normalized the timestamp attribute to a datetime format for easier analysis. We also conducted redundancy and correlation analysis to check for any highly correlated attributes or redundant information, but found that there were no strong correlations or redundancies in the data. There were also no missing values in the dataset after preprocessing, which allowed us to proceed with our analysis without needing to handle missing data. Overall, the data preprocessing steps ensured that we had a clean and well-structured dataset to work with for our exploratory data analysis.

2. Exploratory Data Analysis

2.1. Summary Statistics

2.1.1. Data

2.1.1.1. Unique Counts & Total Listening Events

All-Time

User	Unique Tracks	Unique Artists	Unique Albums	Total Listening Events
Alan	10569	1064	3769	450283
Alexandra	5249	1064	2349	139541
Anthony	9066	1296	4252	72483
Koren	15188	4039	7954	181858
2024				
Alan	3926	798	1442	57042
Alexandra	2177	531	1123	12369
Anthony	2736	551	1442	11713
Koren	3011	1267	1940	14706
2025				
Alan	3227	480	976	29418
Alexandra	2137	503	1064	8712
Anthony	3510	543	1622	13706
Koren	3740	1405	2313	17160

Table 1: Summary Statistics for Each User (All-Time, 2024, 2025)

2.1.1.2. Top Tracks

All-Time		2024		2025	
Track	**Plays**	**Track**	**Plays**	**Track**	**Plays**
4th Dimension	1371	I Remember	122	Begin Again	102
No More Parties In LA	1237	Nosetalgia	117	In Your Eyes	98
SIRENS	1213	Centurion	111	Remember Where You Are	87
Gorgeous	1151	Crosshairs	111	either on or off the drugs	84
The Games We Play	1037	Mykonos	109	The Kill	84
Dark Fantasy	1016	Melt Session #1	107	Soul Control	78
CAN’T SAY	1008	Siberian Breaks	106	Weird Fishes	78
So Appalled	988	Mural	102	Endtroduction	75
Feel The Love	967	Jet Black	101	HAZARD DUTY PAY!	67
Armed And Dangerous	962	Respiration	100	Uncanny long arms	67

Table 2: Alan's Top 10 Tracks (All-Time, 2024, 2025)

All-Time		2024		2025	
Track	**Plays**	**Track**	**Plays**	**Track**	**Plays**
Sugar, We’re Goin Down	237	Sunshine!	49	Vendetta	119
Kali Ma	194	Doubt Me	45	Sunshine!	25
Motion Sickness	189	Might Love Myself	41	Riptide	19
Happy Judgement Day	187	All My Life	34	Heartbreak Of The Century	18
This Is Gospel	184	Too Sweet	31	Static	16

Everybody Talks	181	Riptide	30	Doomed	15
LA Devotee	174	Ronald	28	Fed Up	14
Can't Kick Up The Roots	171	The Drug In Me Is You	27	I've Told You Once	14
Hallelujah	170	The Surface	27	Everybody but You	13
Emperor's New Clothes	168	The Afterglow	26	Nobody Loves Me	13

Table 3: Alexandra's Top 10 Tracks (All-Time, 2024, 2025)

All-Time		2024		2025	
Track	**Plays**	**Track**	**Plays**	**Track**	**Plays**
ball w/o you	234	Not Like Us	59	Yale	51
Father Stretch My Hands Pt. 1	226	one of wun	58	ss	50
I Wonder	190	ILoveUIHateU	50	No Pole	48
Down Below	184	Fried (She a Vibe)	45	POWER	46
Praise The Lord (Da Shine)	179	It's Up	43	tv off	44
Freestyle	172	2 Slippery	40	No Flockin'	40
POWER	171	ALL RED	39	I Wonder	40
Jungle	164	CARNIVAL	38	FIELD TRIP	38
Flashing Lights	160	Cinderella	36	EVIL J0RDAN	38
Shoota	157	Father Stretch My Hands Pt. 1	36	overseas	38

Table 4: Anthony's Top 10 Tracks (All-Time, 2024, 2025)

All-Time		2024		2025	
Track	**Plays**	**Track**	**Plays**	**Track**	**Plays**
Honcho Style 3	297	On the Roof	78	Brown Paper Bag	44
Poles 1469	278	Maximum Style (Lover To Lover)	66	Myron	39
Stacy's Mom	276	Valley of the Shadows	64	Off The Meter	38
From The D To The A	272	Brown Paper Bag	60	Autobots	38
Money Longer	263	Only You (Original Mix)	59	On The Roof	37
First Day Out	262	Rotation	54	Ice Cream & Syrup	36
Lane Changing	258	Ice Cream & Syrup	45	Only You (Original Mix)	35
Award Tour	242	In Too Deep - mixed	44	Maximum Style (Lover To Lover)	32
Anti	235	Pavement	43	Load Out	32
Teflon Flow	226	Limb By Limb - DJ SS Remix	43	Gemini	30

Table 5: Koren's Top 10 Tracks (All-Time, 2024, 2025)

2.1.1.3. Top Artists

All-Time		2024		2025	
Artist	**Plays**	**Artist**	**Plays**	**Artist**	**Plays**
Kanye West	43739	Kanye West	2803	Kanye West	1184

Travis Scott	22269	Kendrick Lamar	2206	JPEGMAFIA	1094
Kendrick Lamar	15485	Danny Brown	1453	Kendrick Lamar	927
Denzel Curry	13157	Freddie Gibbs	1206	Jessie Ware	798
Juice WRLD	9580	Denzel Curry	1189	Denzel Curry	756
Playboi Carti	8932	The Roots	1185	Danny Brown	636
Freddie Gibbs	8815	A Tribe Called Quest	948	Fiona Apple	605
Lil Uzi Vert	8553	Pusha T	943	Freddie Gibbs	585
Drake	8538	Sufjan Stevens	922	JID	585
Future	8149	Gorillaz	875	A Tribe Called Quest	475

Table 6: Alan's Top 10 Artists (All-Time, 2024, 2025)

All-Time		2024		2025	
Artist	**Plays**	**Artist**	**Plays**	**Artist**	**Plays**
Neck Deep	9308	Beartooth	988	Beartooth	563
State Champs	8301	Neck Deep	656	Neck Deep	444
Panic! At The Disco	7117	Bring Me The Horizon	501	State Champs	385
Beartooth	5603	Wage War	442	Wage War	306
Bring Me The Horizon	4320	State Champs	408	Bring Me The Horizon	284
Fall Out Boy	3133	Falling In Reverse	339	Ashnikko	233
blink-182	2697	Ashnikko	316	I Prevail	228
My Chemical Romance	2484	Panic! At The Disco	276	Panic! At The Disco	209
mgk	2461	Kim Dracula	265	WSTR	195
Wage War	2410	mgk	258	mgk	166

Table 7: Alexandra's Top 10 Artists (All-Time, 2024, 2025)

All-Time		2024		2025	
Artist	**Plays**	**Artist**	**Plays**	**Artist**	**Plays**
Kanye West	6780	Kanye West	858	Kanye West	1199
Drake	3829	Future	648	Playboi Carti	640
Future	2132	Kendrick Lamar	428	Drake	557
Young Thug	1959	Drake	426	Young Thug	511
Travis Scott	1457	¥$	335	Future	425
YoungBoy Never Broke Again	1428	Gunna	313	Gunna	342
Kendrick Lamar	1418	21 Savage	301	Kendrick Lamar	321
Playboi Carti	1382	Young Thug	296	Lil Baby	319
21 Savage	1305	YoungBoy Never Broke Again	292	Ken Carson	317
Juice WRLD	1234	BigXthaPlug	280	21 Savage	313

Table 8: Anthony's Top 10 Artists (All-Time, 2024, 2025)

All-Time		2024		2025	
Artist	**Plays**	**Artist**	**Plays**	**Artist**	**Plays**
Lil Uzi Vert	5764	Mint	764	Playboi Carti	1071
Playboi Carti	4323	LTJ Bukem	523	Mint	463
Mint	4034	Lonnie Liston Smith	325	LTJ Bukem	300
G Herbo	2820	Nookie	267	Che	290
Chief Keef	2732	toe	241	Roni Size	279
Cdot Honcho	2654	Alex Reece	231	tenkay	255
Tee Grizzley	2000	Wax Doctor	187	toe	219
Travis Scott	1807	Roni Size	186	Nookie	203
Pi'erre Bourne	1789	Big Bud	165	Lil Uzi Vert	183
Future	1642	Hidden Agenda	162	OsamaSon	181

Table 9: Koren's Top 10 Artists (All-Time, 2024, 2025)

2.1.1.4. Top Albums

All-Time		2024		2025	
Album	**Plays**	**Album**	**Plays**	**Album**	**Plays**
The Life Of Pablo	8732	Undun	687	What's Your Pleasure?	677
My Beautiful...Fantasy	8265	Piñata	677	Melt My Eyez See Your Future	453
Rodeo	7932	Donuts	669	I LAY DOWN MY LIFE FOR YOU	372
ASTROWORLD	7676	Late Registration	665	Wallsocket	359
Die Lit	6645	Melt My Eyez See Your Future	652	I LAY DOWN MY LIFE FOR YOU	356
Goodbye & Good Riddance	6565	Atrocity Exhibition	639	Lianne La Havas	334
Late Registration	6071	To Pimp A Butterfly	589	Since I Left You	331
good kid, m.A.A.d city	5443	Mos Def...Black Star	567	Black On Both Sides	299
KIDS SEE GHOSTS	5086	XXX (Deluxe Version)	550	4eva Is A Mighty Long Time	298
The College Dropout	4976	Bizarre Ride II The Pharcyde	528	Atrocity Exhibition	287

Table 10: Alan's Top 10 Albums (All-Time, 2024, 2025)

All-Time		2024		2025	
Album	**Plays**	**Album**	**Plays**	**Album**	**Plays**
Life's Not Out To Get You	1767	The Surface	322	The Surface	160
Death of a Bachelor	1584	POST HUMAN: NeX GEn	195	Erase The Pain	119
Wishful Thinking	1580	Neck Deep	169	TRUE POWER	116
Tickets To My Downfall	1533	Below	166	Below	112
The Peace And The Panic	1520	A Gradual Decline In Morale	166	Aggressive	107
Around the World and Back	1488	Aggressive	152	Kings of the New Age	83
The Finer Things	1454	TRUE POWER	135	Tickets To My Downfall	81
Around the World and Back	1305	Tickets To My Downfall	129	POST HUMAN: NeX GEn	80

Too Weird to Live, Too Rare to Die!	1282	Disease (Deluxe Edition)	125	Disease (Deluxe Edition)	79
Vices & Virtues	1254	Angel Killer Zero	119	Neck Deep	75

Table 11: Alexandra's Top 10 Albums (All-Time, 2024, 2025)

All-Time		2024		2025	
Album	**Plays**	**Album**	**Plays**	**Album**	**Plays**
Graduation	1240	GNX	274	MUSIC	385
My Beautiful...Fantasy	1051	WE DON'T TRUST YOU	256	Graduation	266
Donda	1022	VULTURES 1	219	My Beautiful...Fantasy	250
The Life Of Pablo	929	Graduation	209	Yeezus	184
Take Care	638	american dream	146	Slime Season	161
Yeezus	546	MIXTAPE PLUTO	131	The Life Of Pablo	156
Watch The Throne	478	My Beautiful...Fantasy	121	A Great Chaos	141
The College Dropout	454	One of Wun	107	GNX	120
Death Race For Love	444	The Life Of Pablo	104	Take Care	113
i am > i was	405	TAKE CARE	103	UY SCUTI	112

Table 12: Anthony's Top 10 Albums (All-Time, 2024, 2025)

All-Time		2024		2025	
Album	**Plays**	**Album**	**Plays**	**Album**	**Plays**
Playboi Carti	1491	Selected Jungle Works	419	MUSIC - SORRY 4 DA WAIT	584
Die Lit	1346	Atmospheric Intelligence	215	MUSIC	307
#SantanaWorld (+)	1165	Producer 01	188	Selected Jungle Works	258
Luv Is Rage	1155	Selected Works 94-96	187	New Forms	211
Lil Uzi Vert vs. The World	1064	Producer 05	161	Atmospheric Intelligence	177
H3	1020	the book about my...vague anxiety	126	REST IN BASS	160
Takeover	1008	Visions of a New World	105	Producer 01	109
Luv Is Rage 2	965	On the Roof / The Flute Tune	101	Selected Works 94-96	93
Underdog	961	Producer 02	88	Unutella	93
The Life Of Pi'erre 4	887	New Forms	86	Slow Motion	90

Table 13: Koren's Top 10 Albums (All-Time, 2024, 2025)

Artist Colors Key

Artist	Color	Artist	Color	Artist	Color
Kanye West		Beartooth		Playboi Carti	
Neck Deep		Kendrick Lamar		Panic! At The Disco	
Future		Denzel Curry		State Champs	
Lil Uzi Vert		Jessie Ware		Drake	

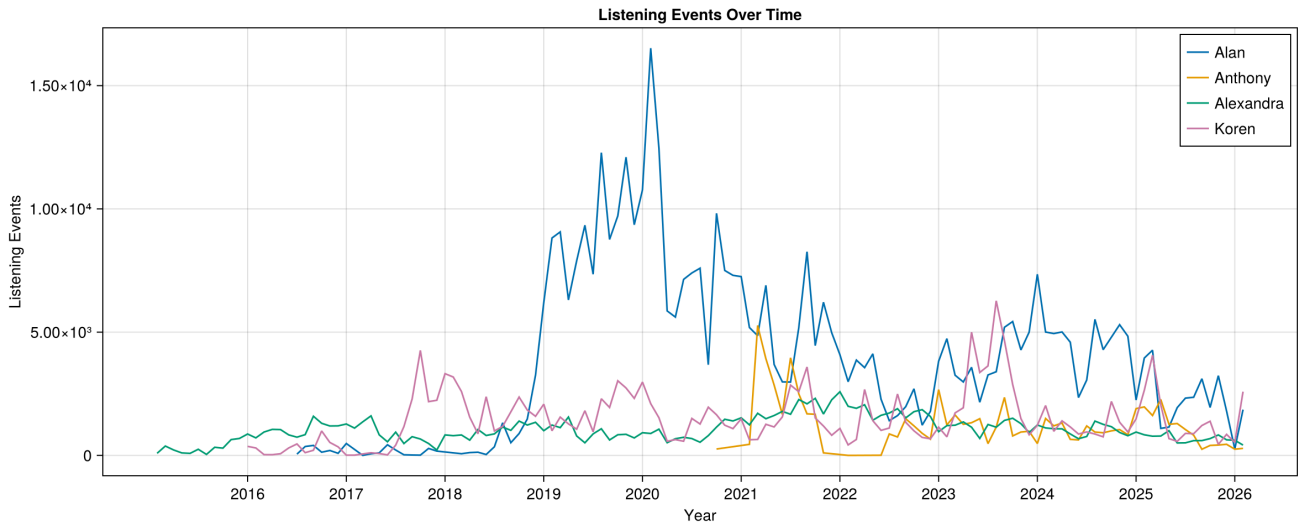
Mint		Travis Scott		21 Savage	
Ken Carson		Roni Size		mgk	
JPEGMAFIA		Juice WRLD		Cdot Honcho	
Danny Brown		Bring Me The Horizon		Young Thug	
LTJ Bukem		Falling In Reverse		I Prevail	
Gunna		¥$		Hidden Agenda	
Nookie		Freddie Gibbs		KIDS SEE GHOSTS	
The Roots		Pusha T		Tee Grizzley	
A Tribe Called Quest		Wage War		toe	
Wax Doctor		Lianne La Havas		Black Star	
underscores		Fall Out Boy		Palisades	
Lil Baby		Tom & Jerry		Remarc	
SOB X RBE		Duwap Kaine		J Majik	
Ashnikko		Kim Dracula		YoungBoy Never Broke Again	
BigXthaPlug		Lonnie Liston Smith		Che	
Pi'erre Bourne		Earl Sweatshirt		DANGERDOOM	
Fleet Foxes		MGMT		Lupe Fiasco	
Anderson .Paak		Hozier		Sleep Theory	
Neon Trees		Woe, Is Me		Silverstein	
Rivals		Don Toliver		Roddy Ricch	
A$AP Rocky		Luh Tyler		Kodak Black	
A Boogie Wit da Hoodie		Trippie Redd		Fountains Of Wayne	
Origin Unknown		ark762		Total Science	
JMJ		Benji Blue Bills		Cutty Ranks	
Fiona Apple		Sufjan Stevens		JID	
Gorillaz		blink-182		My Chemical Romance	
WSTR		G Herbo		Chief Keef	
Alex Reese		tenkay		Big Bud	
OsamaSon		J Dilla		The Avalanches	
Mos Def		Big K.R.I.T.		The Pharcyde	
Diamond Construct		JAY-Z		Tay-K	
PFM		Nuito			

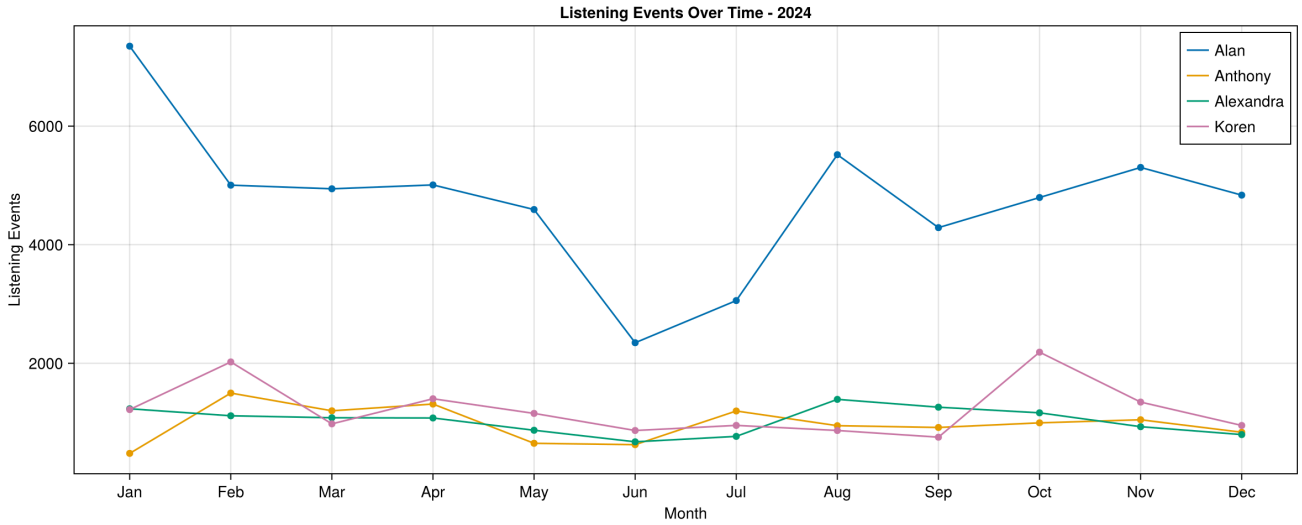
2.1.2. Analysis

2.1.3. Methodology

2.2. Events Over Time

2.2.1. Data





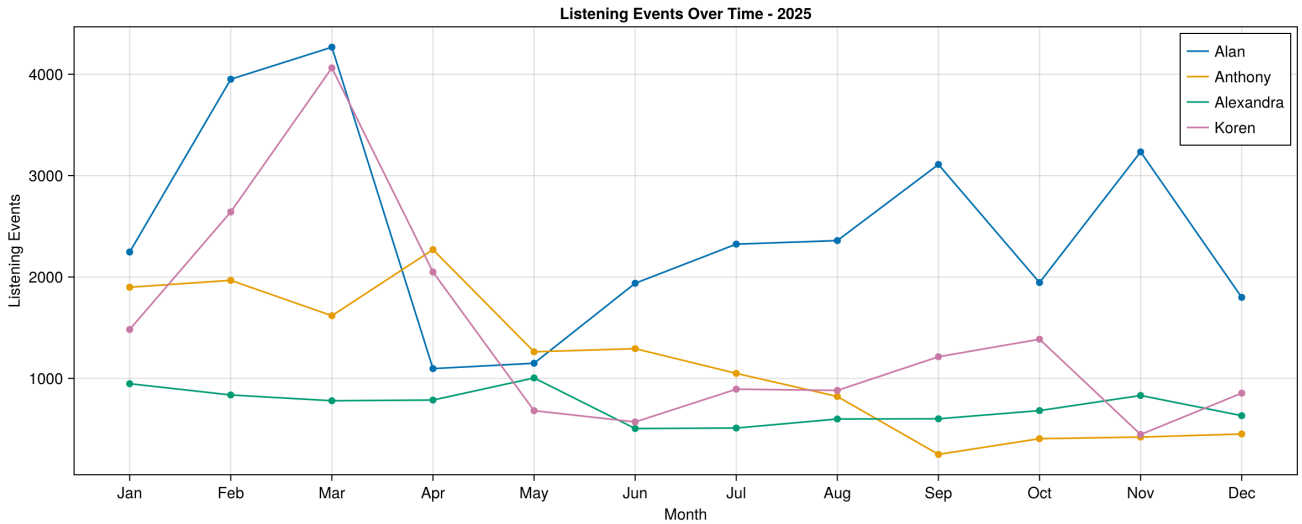


Figure 1: Daily Listening Events Over Time for Each User (All-Time, 2024, 2025)

2.2.2. Analysis

2.2.3. Methodology

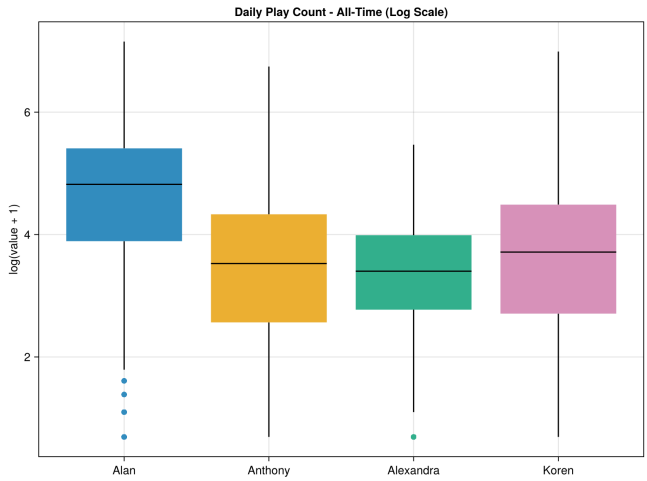
2.3. Distributions

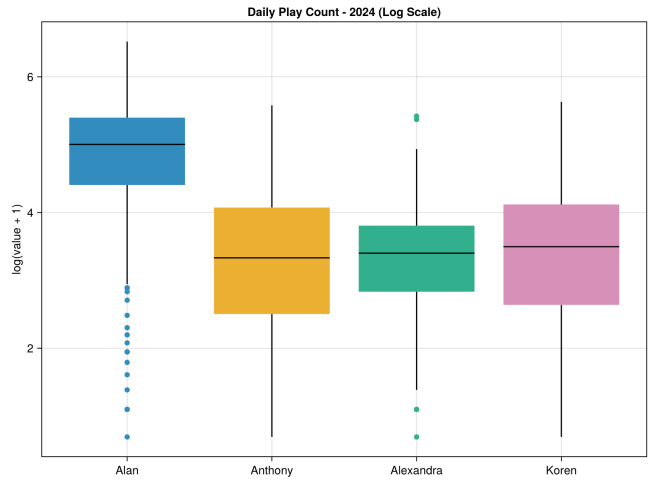
2.3.1. Data

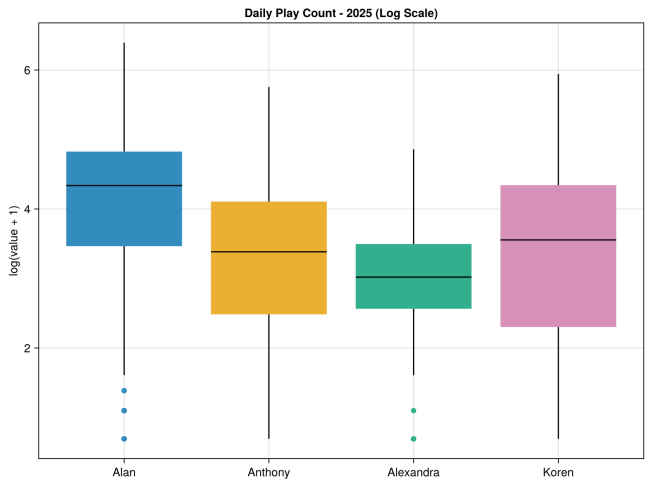
2.3.1.1. Daily Play Count

All-Time							
	Mean	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	156.57	123.00	1.00	1276.00	144.12	20771.13	1.87
Alexandra	39.78	29.00	12.00	235.00	34.62	1198.52	1.71
Anthony	58.08	33.00	1.00	848.00	73.25	5365.18	3.17
Koren	63.92	40.00	1.00	1084.00	74.14	5496.12	2.91
2024							
Alan	160.68	148.00	165.00	676.00	105.76	11184.63	1.02
Alexandra	34.26	29.00	9.00	224.00	27.36	748.46	2.49
Anthony	42.13	27.00	1.00	263.00	43.09	1856.87	1.89
Koren	43.38	32.00	1.00	277.00	41.99	1763.30	2.10
2025							
Alan	54.82	75.50	2.00	596.00	71.52	5115.28	1.91
Alexandra	24.47	19.50	12.00	127.00	17.56	308.41	1.74
Anthony	45.38	28.50	6.00	314.00	51.83	2686.58	2.43
Koren	54.82	34.00	1.00	379.00	60.89	3707.91	1.91

Table 14: Distribution of Daily Play Counts for Each User (All-Time, 2024, 2025)







2.3.1.2. Plays Per Track

All-Time							
	Mean	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	47.52	5.00	2.00	1370.00	119.01	14164.13	4.16
Alexandra	32.20	20.00	2.00	262.00	37.28	1389.89	1.75
Anthony	8.78	3.00	2.00	233.00	16.41	269.32	5.03
Koren	13.18	4.00	2.00	305.00	23.60	557.08	4.22
2024							
Alan	15.01	4.00	1.00	121.00	20.35	414.10	1.63
Alexandra	6.06	5.00	4.00	51.00	4.60	21.15	2.85
Anthony	4.42	2.00	1.00	58.00	6.14	37.76	3.41
Koren	5.00	2.00	1.00	77.00	7.49	56.13	3.30
2025							
Alan	9.47	4.00	1.00	103.00	11.51	132.40	2.39
Alexandra	4.47	4.00	3.00	118.00	3.64	13.23	16.46
Anthony	4.06	2.00	1.00	50.00	5.47	29.87	3.50

Koren	4.70	2.00	1.00	44.00	5.73	32.84	2.23

Table 15: Distribution of Plays per Track for Each User (All-Time, 2024, 2025)

2.3.1.3. Plays Per Artist

All-Time							
	Mean	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	265.65	5.00	2.00	43738.00	1547.60	2395064.73	16.81
Alexandra	131.15	22.00	2.00	9307.00	557.96	311323.54	11.00
Anthony	55.93	4.00	2.00	6779.00	273.06	74559.34	15.17
Koren	45.03	6.00	2.00	5763.00	198.37	39350.67	14.55
2024							
Alan	71.48	5.00	2.00	2802.00	211.32	44655.85	6.09
Alexandra	23.29	7.00	1.00	987.00	70.99	5039.69	8.07
Anthony	21.26	3.00	1.00	857.00	65.88	4340.80	7.26
Koren	11.61	3.00	1.00	763.00	35.68	1272.94	11.74
2025							
Alan	61.29	8.00	1.00	1183.00	135.04	18235.16	4.34
Alexandra	17.32	5.00	3.00	562.00	47.73	2278.42	6.93
Anthony	25.24	4.00	1.00	1198.00	81.50	6642.28	8.19
Koren	12.21	3.00	1.00	1070.00	40.27	1621.82	15.49

Table 16: Distribution of Plays per Artist for Each User (All-Time, 2024, 2025)

2.3.1.4. Plays Per Album

All-Time							
	Mean	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	114.75	4.00	2.00	8731.00	503.24	253250.73	9.14
Alexandra	57.97	18.00	2.00	1766.00	150.36	22609.62	6.19
Anthony	16.64	4.00	2.00	1239.00	52.37	2742.38	10.64
Koren	21.64	5.00	2.00	1490.00	61.03	3724.59	10.40
2024							
Alan	38.28	4.00	2.00	686.00	92.17	8494.98	3.78
Alexandra	10.91	6.00	1.00	321.00	20.73	429.59	6.63
Anthony	8.07	3.00	1.00	273.00	18.28	334.03	7.64
Koren	7.21	2.00	1.00	418.00	16.57	274.53	11.38
2025							
Alan	29.66	5.00	1.00	676.00	58.62	3436.35	4.00
Alexandra	8.11	5.00	2.00	159.00	13.69	187.32	5.03
Anthony	8.30	3.00	1.00	384.00	20.01	400.26	8.69
Koren	7.12	2.00	1.00	583.00	18.34	336.26	17.53

Table 17: Distribution of Plays per Album for Each User (All-Time, 2024, 2025)

2.3.1.5. Session Length

All-Time							
	Mean	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	41.36	26.50	1.00	1158.65	51.30	2631.50	4.09
Alexandra	30.54	19.06	1.00	494.45	39.98	1598.04	3.78
Anthony	44.01	21.21	2.00	1199.27	67.80	4596.43	5.25
Koren	30.46	13.01	1.00	1085.80	52.12	2716.85	5.97
2024							
Alan	58.19	36.44	1.00	1158.64	81.48	6638.73	5.12
Alexandra	28.37	17.27	3.00	494.45	41.44	1717.48	4.36
Anthony	39.27	19.95	1.00	435.77	53.36	2846.83	3.26
Koren	42.02	19.35	1.00	711.71	64.38	4145.35	3.56
2025							
Alan	42.87	27.97	1.00	678.58	55.24	3050.96	4.08
Alexandra	21.77	15.70	7.00	260.91	24.86	617.96	3.43
Anthony	49.83	24.18	3.00	1199.27	90.70	8226.07	6.29
Koren	50.69	20.59	1.00	877.88	82.47	6800.97	4.18

Table 18: Distribution of Session Lengths for Each User (All-Time, 2024, 2025)

2.3.2. Analysis

2.3.3. Methodology

2.4. Outliers

2.4.1. Data

2.4.1.1. Daily Play Count

All-Time								
	Outliers	**Mean**	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	94	140.48	118.00	1.00	479.00	112.43	12640.24	0.82
Alexandra	191	34.01	28.00	12.00	109.00	24.83	616.58	0.97
Anthony	89	42.61	30.00	1.00	168.00	40.15	1611.64	1.15
Koren	158	50.92	37.00	1.00	198.00	47.00	2208.74	1.08
2024								
Alan	5	155.13	146.50	165.00	427.00	95.29	9079.38	0.52
Alexandra	17	30.21	28.00	9.00	83.00	18.83	354.56	0.67
Anthony	14	35.28	25.00	1.00	125.00	30.72	943.94	1.08
Koren	17	36.70	30.00	1.00	130.00	29.10	846.94	0.85
2025								

Alan	5	81.58	72.00	2.00	249.00	60.15	3618.18	0.63
Alexandra	15	22.07	18.00	12.00	58.00	13.19	173.89	0.79
Anthony	20	34.59	26.50	6.00	130.00	29.84	890.66	1.08
Koren	17	44.61	31.00	1.00	166.00	42.62	1816.50	1.04

Table 19: Distribution of Daily Play Counts for Each User with Outliers Removed (All-Time, 2024, 2025)

2.4.1.2. Plays Per Track

All-Time								
	Outliers	**Mean**	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	1736	7.12	3.00	2.00	48.00	9.29	86.22	2.46
Alexandra	281	25.13	17.00	2.00	103.00	25.82	666.52	1.20
Anthony	1004	4.19	3.00	2.00	16.00	3.71	13.75	1.62
Koren	1523	6.58	3.00	2.00	31.00	7.14	51.05	1.70
2024								
Alan	146	12.62	3.00	1.00	58.00	16.54	273.70	1.37
Alexandra	68	5.51	5.00	4.00	14.00	3.26	10.62	0.69
Anthony	231	2.84	2.00	1.00	10.00	2.38	5.66	1.64
Koren	345	2.65	2.00	1.00	10.00	2.28	5.22	1.79
2025								
Alan	148	7.65	4.00	1.00	31.00	7.77	60.34	1.13
Alexandra	51	4.18	4.00	3.00	9.00	2.14	4.58	0.53
Anthony	404	2.43	2.00	1.00	7.00	1.76	3.09	1.41
Koren	344	3.26	2.00	1.00	12.00	3.30	10.92	1.63

Table 20: Distribution of Plays per Track for Each User with Outliers Removed (All-Time, 2024, 2025)

2.4.1.3. Plays Per Artist

All-Time								
	Outliers	**Mean**	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	320	7.51	3.00	2.00	55.00	10.28	105.59	2.59
Alexandra	121	33.02	17.00	2.00	180.00	39.93	1594.69	1.57
Anthony	207	6.87	3.00	2.00	41.00	8.51	72.40	2.09
Koren	521	10.46	4.50	2.00	56.00	12.82	164.38	1.86
2024								
Alan	138	8.12	3.00	2.00	61.00	11.83	139.93	2.47
Alexandra	74	7.19	6.00	1.00	25.00	5.23	27.38	1.16
Anthony	76	5.08	3.00	1.00	25.00	5.72	32.76	1.88
Koren	184	3.70	2.00	1.00	17.00	3.63	13.15	1.83
2025								
Alan	70	18.67	5.00	1.00	113.00	27.09	733.92	1.88

Alexandra	78	5.50	5.00	3.00	18.00	3.61	13.03	1.33
Anthony	71	6.52	3.00	1.00	33.00	7.52	56.54	1.72
Koren	128	5.08	2.00	1.00	25.00	5.78	33.39	1.68

Table 21: Distribution of Plays per Artist for Each User with Outliers Removed (All-Time, 2024, 2025)

2.4.1.4. Plays Per Album

All-Time								
	Outliers	**Mean**	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	753	5.76	3.00	2.00	38.00	6.84	46.79	2.41
Alexandra	217	24.90	15.00	2.00	117.00	27.78	771.71	1.41
Anthony	618	4.78	3.00	2.00	21.00	4.57	20.88	1.75
Koren	1002	8.33	4.00	2.00	41.00	9.55	91.27	1.71
2024								
Alan	255	6.76	3.00	2.00	46.00	9.71	94.36	2.43
Alexandra	103	6.10	5.00	1.00	19.00	4.05	16.42	0.90
Anthony	170	3.68	2.00	1.00	15.00	3.58	12.80	1.69
Koren	292	3.06	2.00	1.00	12.00	2.73	7.43	1.74
2025								
Alan	142	10.56	3.00	1.00	67.00	14.63	214.01	1.97
Alexandra	108	4.63	4.00	2.00	12.00	2.66	7.10	0.86
Anthony	180	3.64	2.00	1.00	15.00	3.42	11.70	1.71
Koren	138	4.66	2.00	1.00	20.00	5.04	25.42	1.43

Table 22: Distribution of Plays per Album for Each User with Outliers Removed (All-Time, 2024, 2025)

2.4.1.5. Session Length

All-Time								
	Outliers	**Mean**	**Median**	**Mode**	**Range**	**Standard Deviation**	**Variance**	**Skewness**
Alan	798	32.95	24.14	1.00	133.23	31.60	998.39	1.06
Alexandra	922	21.04	17.27	1.00	75.61	16.85	283.85	0.97
Anthony	178	31.43	18.48	2.00	140.12	32.92	1083.91	1.18
Koren	796	19.58	10.81	1.00	91.81	21.93	480.72	1.30
2024								
Alan	103	43.47	33.20	1.00	168.25	40.73	1658.75	1.07
Alexandra	77	20.05	15.20	3.00	76.48	17.63	310.72	1.02
Anthony	31	29.84	18.07	1.00	132.36	29.94	896.46	1.09
Koren	100	25.98	15.65	1.00	118.22	27.52	757.41	1.27
2025								
Alan	90	32.44	24.89	1.00	128.02	30.39	923.71	1.02
Alexandra	70	17.15	14.26	7.00	59.03	13.59	184.57	0.90

Anthony	65	30.70	20.88	3.00	125.17	29.44	866.96	1.27
Koren	65	34.33	16.78	1.00	157.38	39.37	1549.89	1.28

Table 23: Distribution of Session Lengths for Each User with Outliers Removed (All-Time, 2024, 2025)

2.4.2. Analysis

2.4.3. Methodology

2.5.

Exploratory Data Analysis

1. outliers
   - extreme values (for any/all attributes)
   - irregular data points
   - visualize them
   - z-score?
   - iqr
2. univariate analysis
   - distributions
   - skewness
   - outliers
   - histograms
   - boxplots
   - q-q plots
   - counts, unique counts
   - ranges, sd, mean, median, mode
   - 5 num
3. bivariate analysis &/or
4. multivariate analysis
   - scatterplots - numeric pairs
   - bar plots - categorical
   - correlations - matrix
   - group comparisons - differences across categories - pivot tables
   - distributions
   - bubble chart
   - heat map
   - run/line chart
   - pair plots
   - pca?
   - time series analysis
5. K-means clustering
6. transformation?
   - scaling?
   - encoding?
7. simple predictive models
   - linear/multiple linear regression
8. visualizing insights

9. patterns
10. anomalies

Exploratory Data Analysis

Summary stats - done

dataset info

- approx 872k listening events
- 4 people
- time span: 2015-2026 (full overlap from 2020-2026)
- no missing values

```
Summary Statistics

Group Statistics

Total listening events: 844165
Unique tracks: 30717
Unique artists: 5929
Unique albums: 13533
Unique platforms: 275
Unique countries: 19

Individual Statistics

Alan:
  Total events: 450283
  Unique tracks: 10569
  Unique artists: 1695
  Unique albums: 3769
  Unique platforms: 48
  Unique countries: 16
Alexandra:
  Total events: 139541
  Unique tracks: 5249
  Unique artists: 1064
  Unique albums: 2349
  Unique platforms: 75
  Unique countries: 4
Anthony:
  Total events: 72483
  Unique tracks: 9066
  Unique artists: 1296
  Unique albums: 4252
  Unique platforms: 24
  Unique countries: 2
Koren:
  Total events: 181858
  Unique tracks: 15188
  Unique artists: 4039
  Unique albums: 7954
  Unique platforms: 143
  Unique countries: 5
```

- summary_statistics.jl
- simple sql queries
- count rows

- insights:
  - lot of events - 11 years of data
  - lots of unique tracks, artists, albums, platforms, countries
  - platforms? - they are mostly different versions of ios (iphone, ipad, diff updates) & android (diff updates, phones, tablets) - possible to see exactly what device was used to listen and when
    - cast, home, web player, smart tv, win 10, 11, macos, linux
  - countries:
    - koren - us, ca (correct, mostly us) - also gb and switzerland - prob vpn
    - alan - many - either travel or vpn
  - users have different listening habits - some listen to more unique tracks, artists, albums, platforms, countries than others
  - top listener by raw numbers: alan - 450k events
  - top listener by unique tracks: koren - 15k unique tracks
  - top listener by unique artists: koren - 4k unique artists
  - top listener by unique albums: koren - 7k unique albums
  - top listener by unique platforms: koren - 143 unique platforms
  - include runners up here and do comparisons
  - insights about listening habits - alan listens to a lot of music but not as much variety as me
  - i listen to a lot of music with a lot of variety
    - 181k events but 15k unique tracks, 4k unique artists, 7k unique albums, 143 unique platforms, 5 unique countries
  - alan - has fav artists and albums - listens to them a lot - less variety - but still wins by raw numbers
    - 450k events but only 10k unique tracks, 1.7k unique artists, 3.7k unique albums, 48 unique platforms, 16 unique countries
  - others:
    - anthony - 72k events but only 9k unique tracks, 1.3k unique artists, 4.2k unique albums, 24 unique platforms, 2 unique countries
    - alexandra - 139k events but only 5k unique tracks, 1k unique artists, 2.3k unique albums, 75 unique platforms, 4 unique countries
  - lowest listener by raw numbers: anthony - 72k events
  - lowest listener by unique tracks: alexandra - 5k unique tracks
  - lowest listener by unique artists: alexandra - 1k unique artists - also anthony - 1.3k unique artists - they have fav artists they listen to a lot but not as much variety
  - lowest listener by unique albums: alexandra - 2.3k unique albums
  - lowest listener by unique platforms: anthony - 24 unique platforms
  - lowest listener by unique countries: anthony - 2 unique countries - only a few devices, and doesnt use vpn or travel much
  - overall - lot of variety in listening habits - some people listen to a lot of music but not as much variety, while others listen to a lot of music with a lot of variety - keep in mind tho 11 years

  follow up:

  - more detailed analysis of listening habits - what artists, albums, platforms, countries do they listen to the most? - top 10 artists, albums, platforms, countries for each user - compare and contrast
  - also look at temporal patterns - when do they listen to music? - time of day, day of week, month, year - do they have different listening habits at different times? - do they listen to more music on weekends or weekdays? - do they listen to more music in certain months or years? - do they have different listening habits at different times of day? - do they listen to more music in the morning, afternoon, evening, night?

  - also look at behavioral patterns - do they skip songs a lot? - do they listen to a lot of music offline? - do they shuffle a lot? - do they have different listening habits when they are offline vs online? - do they have different listening habits when they are shuffling vs not shuffling? - do they have different listening habits when they are skipping vs not skipping?
  - also look at group-level comparisons - do they have similar listening habits? - do they listen to the same artists, albums, platforms, countries? - do they have similar temporal patterns? - do they have similar behavioral patterns? - do they have similar listening habits overall?
  - also look at dimensional reduction and early unsupervised structure - can we find any patterns in the data? - can we cluster users based on their listening habits? - can we find any latent factors that explain their listening habits? - can we find any interesting patterns in the data that we didnt expect?
  - also look at preliminary predictive framing - can we predict how much music they will listen to in the future? - can we predict what artists, albums, platforms, countries they will listen to in the future? - can we predict when they will listen to music in the future? - can we predict their behavioral patterns in the future? - can we predict their group-level comparisons in the future? - can we predict their dimensional reduction and early unsupervised structure in the future?
  - also story telling - over time

summary stats: visualizations

- bar chart per user for:
  - total listening events - total_listening_events_per_user.png
  - unique tracks - unique_tracks_per_user.png
  - unique artists - unique_artists_per_user.png
  - unique albums - unique_albums_per_user.png
- ratio plots for:
  - unique tracks / total listening events
- for timestamp vs listening events:
  - line chart for each user over time - events_over_time.png
- listening clock - listening_times_name.png - clock plot for each user showing when they listen to music
- explain all the functions & methodology

Univariate analysis - done

- averages - group and per user:

```
Group Averages

Average ms_played: 75742.9187
Average skip rate: 0.2466
Average shuffle rate: 0.6784
Average offline rate: 0.0112
Average daily plays: 370.6499
Average daily unique tracks: 256.3604
Average daily unique artists: 111.6732
Average monthly plays: 9245.0903
Average yearly plays: 103819.4117

User Averages

User: Alan
  Average ms_played: 59069.6243
  Average skip rate: 0.2598
  Average shuffle rate: 0.6332
```

```
  Average offline rate: 0.0013
  Average daily plays: 289.1867
  Average daily unique tracks: 170.2235
  Average daily unique artists: 51.7361
  Average monthly plays: 6738.8346
  Average yearly plays: 73728.5447

User: Alexandra
  Average ms_played: 125910.0511
  Average skip rate: 0.1175
  Average shuffle rate: 0.9329
  Average offline rate: 0.0498
  Average daily plays: 69.8994
  Average daily unique tracks: 62.4683
  Average daily unique artists: 36.378
  Average monthly plays: 1285.7449
  Average yearly plays: 14565.6158

User: Anthony
  Average ms_played: 91987.2519
  Average skip rate: 0.3411
  Average shuffle rate: 0.5382
  Average offline rate: 0.004
  Average daily plays: 150.382
  Average daily unique tracks: 129.8723
  Average daily unique artists: 55.8233
  Average monthly plays: 2042.6474
  Average yearly plays: 16638.1659

User: Koren
  Average ms_played: 72058.1274
  Average skip rate: 0.2753
  Average shuffle rate: 0.6512
  Average offline rate: 0.0088
  Average daily plays: 149.8734
  Average daily unique tracks: 107.9461
  Average daily unique artists: 63.8412
  Average monthly plays: 2329.4371
  Average yearly plays: 20646.4081
```

- average session rate - group and per user

```
Average session length (minutes): 35.6912

Average session length per user (minutes)

User: Alan
  Average session length: 41.358
User: Alexandra
  Average session length: 30.5397
User: Anthony
  Average session length: 44.0051
User: Koren
  Average session length: 30.4578
```

- methodology:
  - session length - 30 mins
  - if gap between two consecutive tracks > 30 mins - new session

  - assign session boundaries based on this - time diff between each track and prev one, per user
  - lag() - previous track's timestamp - mark as new session start
  - use cumulative sum of those session-start flags - assign "session id"
  - aggregate sessions to get their duration (max(timestamp) - min(timestamp)) per session
  - average those durations

derived attributes

- plays per artist (top 100) + per user
- plays per track (top 100) + per user
- plays per album (top 100) + per user
- insights...

Univariate analysis - Per year

- plays per artist (top 100) + per user
- plays per track (top 100) + per user
- plays per album (top 100) + per user
- insights...
- explain all the functions & methodology

OUTLIERS

- for each user and each day (all time + yearly to compare):
  - daily play count
- mean & sd of daily play count for a user
- First apply log transformation
  - log_play_count = log(1 + track_play_count)
- z = (val - mean) / sd
- outlier -> |z| > 3
- if symmetric - both high and low outliers
- 1.5iqr - prob better
- iqr = q3 - q1 of DPC
- q1 - 1.5 x iqr and q3 + 1.5 x iqr = bounds
- outside = outlier
- can use boxplot to show this too
- Rank-frequency plot - Zipf style
- Pareto chart
- lorenz curve
- histogram of log counts
- same thing for
  - plays per track
  - plays per artist
  - session length

DISTRIBUTIONS

- Daily Listening Counts
  - symmetric?
  - right-skewed?
  - long tail?
  - typical daily listening volume
- daily plot count per user
- plot histogram 50 bins
- overlay density curve
- compute mean, median, sd, skewness, kurtosis
- boxplot
- log scale histogram
- median diff from mean?
- one user has higher variance?
- multiple modes?
- and do this for
  - plays per track
  - plays per artist
  - session length

Bivariate analysis

Multivariate analysis

Temporal analysis

Behavioral pattern exploration

Group-level comparison analysis

Dimensional reduction and early unsupervised structure

Preliminary predictive framing

Story construction